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Research Paper

Does the revised LDCT lung cancer screening guideline bridge the racial disparities gap: Results from the health and retirement study

Chien-Ching Li [a], Jason Manella [b], Safa El Kefi [c], Alicia K. Matthews [c] Show more

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Abstract

Purpose

This study examined racial/ethnic disparities in lung cancer screening eligibility rates using 2013 US Preventive Services Task Force (USPSTF) guidelines for lung cancer with low-dose computed tomography (LDCT) and the revised 2021 guidelines.

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The study utilized a retrospective and cross-sectional research design by analyzing data from the Health and Retirement Study (HRS). N = 2,823 respondents aged 50–80 who self-reported current

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smoking were included in the analyses. Binary logistic regression analysis was conducted to examine the changed status of LDCT screening eligibility based on the revised 2021 guidelines by race/ethnicity after adjusting for respondent demographics.

Results

Our study found substantial increases in screening eligibility rates across racial and ethnic groups when comparing the original and revised guidelines. The largest increase was observed among Black people (174%), Hispanics (152%), those in the other category (118%), and Whites who smoke (80.8%). When comparing original screening guidelines to revised guidelines, Whites who smoke had the highest percentage of changes from “not eligible” to “eligible” (28.3%), followed by individuals in the ”other” category (28.1%), Black people (23.2%) and Hispanics who smoke (18.3%) ($p<0.001$). Binary logistic regression results further showed that Black people who smoke (OR = 0.71, $p=0.001$), as well as Hispanics who smoke (OR=0.54, $p<0.001$), were less likely to change from not eligible to eligible for screening compared to Whites who smoke after adopting the revised screening guidelines. Based on the absolute differences in screening eligibility rates between Whites and other racial/ethnic groups, the disparities may have widened under the new guidelines, particularly with larger absolute differences observed between Whites, Black people, and Hispanics.

Conclusions

Our study highlights racial/ethnic disparities in LDCT screening eligibility among people who currently smoke. While the revised USPSTF guidelines increased screening eligibility for racial and ethnic minorities, they did not eliminate these disparities and may have widened under the new guidelines. Targeted interventions and policies are necessary to address barriers faced by underrepresented populations and promote equitable access to lung cancer screening.

Keywords

Lung cancer screening; Uscreening guidelines; Race/ethnicity; Disparities

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In the United States (US), lung cancer accounts for the highest number of cancer-related deaths.[1,2] More than 349 people are projected to die daily from lung cancer in 2023, more than the combined deaths from breast, prostate, and pancreatic cancers.[1,2] A large and consistent body of research suggests differences in lung cancer incidence and mortality based on race/ethnicity,[3,4] with the highest rates observed among African American men. [2,5] African American men are about 12% more likely to develop lung cancer than White men 5 and 9.8% more likely to die from the disease.[6]

The higher incidence of lung cancer among African Americans can be attributed to genetic, behavioral, environmental, and socioeconomic factors.[7, 8, 9, 10] For instance, African Americans are less successful in quitting smoking compared to other ethnic subgroups,[11] which increases their risk of developing the disease. Also, African American people who smoke have greater exposure to tobacco carcinogens at lower amounts of cigarette consumption than Whites and Hispanics who smoke due to increased intensity of smoking and slower nicotine metabolism and clearance.[12,13] The universal health benefits of smoking cessation have been repeatedly emphasized.[14] However, underrepresented racial/ethnic groups are less likely to receive tobacco counseling and be offered smoking cessation alternatives from providers than Whites who smoke, especially if uninsured.[14] Factors such as lower income, limited education, and reduced access to quality care can impact the likelihood of smoking among African Americans, contributing to the higher incidence of lung cancer among them.[8,12,15]

Environmental factors also confer an increased risk of developing lung cancer. Occupational and environmental exposures are more common in underrepresented minority populations and those with lower socioeconomic status, potentially contributing to existing disparities in lung cancer incidence.[12,16] In addition, genetic factors may also be responsible, as a study found that certain genetic variations in African Americans could increase their cancer susceptibility.[17] Genetic factors can influence an individual's ability to metabolize carcinogens in tobacco smoke, potentially creating disparities in the vulnerability to the harmful effects of smoking among different people.[17] While several studies have proposed a link between African ancestry and lung cancer outcomes, other research has indicated that social determinants of health (e.g., access to healthcare) play a more significant role in shaping these outcomes.[18,19]

Lung cancer early detection screening significantly reduces mortality rates among adults with a history of chronic high frequency smoking.[20] The US Preventive Services Task Force's (USPSTF) recommendation of annual LDCT lung cancer screening (in adults aged 55–80 years who have a 30-pack-year smoking history, currently smoke, or have quit within the past 15 years) was informed by the National Lung Screening Trial (NLST).[21,22] However, studies have shown that the previous

guidelines did not account for race and sex differences in smoking patterns and lung cancer burden.[23, 24, 25] For example, a large and consistent body of research suggests that on average African Americans smoke fewer cigarettes per day than other groups.[26, 27, 28] Due to the increased likelihood of being light people who smoke, emerging research suggests that African American people who smoke are less likely to meet pack-year requirements for lung cancer screening eligibility compared to Whites who smoke.[29] Systematic differences in eligibility for lung cancer screening based on race/ethnicity have the potential to exacerbate existing lung cancer disparities and represent a significant challenge to the appropriateness of established screening guidelines.[30, 31, 32]

To address these issues, the USPSTF expanded the screening eligibility criteria in March 2021 by lowering the minimum age from 55 to 50 years and decreasing the smoking intensity from 30 to 20 pack-years.[23] While the expanded criteria are expected to increase the number of high-risk individuals eligible for screening and reduce lung cancer mortality, the impact on racial and ethnic minorities, including African Americans, has shown mixed findings.[33, 34, 35] Further research is needed to assess the population-level outcomes associated with the revised guidelines and their impact on reducing disparities in lung cancer screening.

The primary objective of this study was to examine the disparities in screening eligibility for lung cancer screening with LDCT based on race and ethnicity. Specifically, the study aimed to compare the eligibility rates using the revised 2021 USPSTF guidelines with the original guidelines from 2013. A secondary goal of the study was to assess whether the revisions made to the guidelines effectively reduced the observed disparities in screening eligibility rates between people who smoke of color and Whites who smoke.

Methods

Study Sample and Data Source

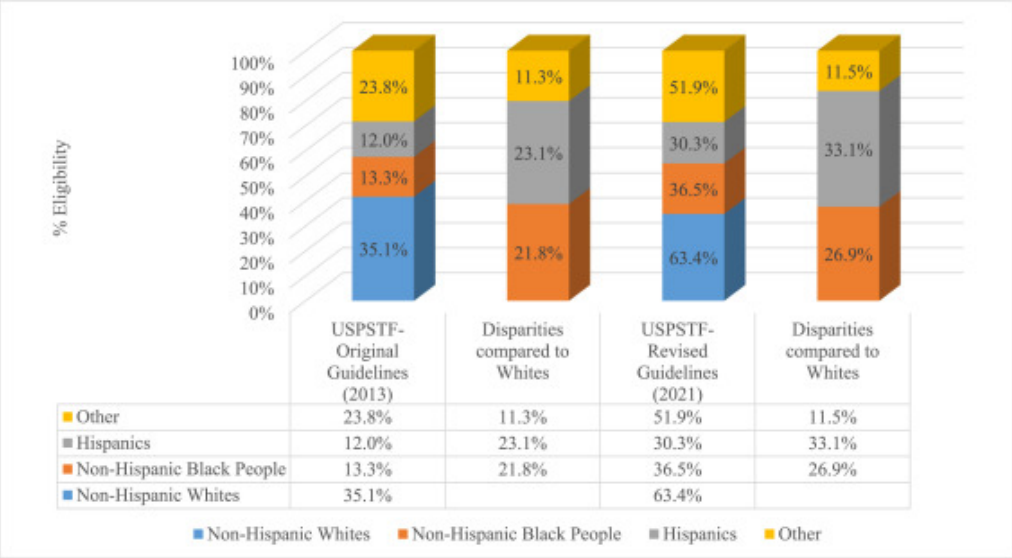
This study utilized a retrospective and cross-sectional research design by analyzing the 2016 Health and Retirement Study (HRS) data. The HRS, conducted by the University of Michigan since, is an ongoing longitudinal study that has used a multi-stage area probability design to collect a wide range of data such as income and wealth, employment, psychological and lifestyle factors, health and health services use from a nationally representative sample of community-dwelling adults aged 51 years and older in the United States.[36] Respondents who self-reported as people who currently smoke aged between 50 and 80 were analyzed.

Measures

The dependent variable of interest was changes in eligibility status for LDCT lung cancer screening (changed, unchanged) among respondents based on the revised USPSTF screening guidelines. According to original guidelines, people who met the following criteria were eligible for screening, including (1) age between 55 and 80, (2) individuals who smoke or those who quit smoking within the past 15 years (smoking was defined as individuals who smoke cigarettes), and (3) with a 30 pack-years smoking history (the number of pack years smoked is calculated by multiplying the number of average packs of cigarettes smoked per day by the number of years the person has smoked). The revised USPSTF guideline[23] lowered the age limit to 50 years and the required number of pack years from 30 to 20. Based on the revised guidelines, people who became eligible for screening were coded as changed in eligibility status. Those still eligible or not eligible for screening were coded as unchanged. The independent variable was race and ethnicity, including the following groups: (1) non-Hispanic Whites, (2) non-Hispanic Black people, (3) Hispanics, and (4) other (i.e., American Indian, Alaskan Native, Asian, Native Hawaiian, and Pacific Islander). Other covariates included age (years), sex (male, female), marital status (married, separated/divorced/widowed, never married), education (high school and below, some college, college and above), household income (<25th percentile, 25-50th percentile, 50-75th percentile, >75th percentile), insurance status (insured, not insured), and self-rated health (excellent, very good, good, fair, poor).

Statistical Analysis

Descriptive statistics (frequency, percentage, mean, standard deviation) were used to describe the characteristics of the study sample. Further, the following rates were calculated to assess whether disparities worsen in screening eligibility rates after adopting revised USPSTF guidelines, including: (1) percentage increase in eligibility: measuring the percentage increase in the number of screening-eligible individuals within each race/ethnic group (i.e., % Increase in Eligibility; Table 2); (2) Percentage changes in eligibility: measuring the percentage increase in screening-eligible rates within each race/ethnic group (i.e., % Changes in Eligibility; Table 2); and (3) Absolute differences in screening eligibility rates between people of color and Whites (i.e., Disparities compared to Whites; Figure 1). Bivariate statistics (independent t-test, Chi-Square test) were used to examine the associations between dependent and independent variables and covariates. Binary logistic regression was conducted to examine the association between dependent and independent variables after adjusting for demographic variables (age, sex, marital status, education, household income, insurance, self-reported health).



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Figure 1. Racial and Ethnic Disparities in USPSTF Screening Eligibility

Results

Characteristics of Study Sample

The study included a sample of 2,823 people who currently smoke aged 50-80 who participated in the 2016 Health and Retirement Study. The mean age of the participants was 60.7 years (SD = 8.2). Most sample consisted of female participants (54.3%) and non-Hispanic Whites (50.6%). Nearly half of the sample (47.6%) reported being separated, divorced, or widowed. The majority of the participants had insurance coverage (84.9%), reported good health or above (self-rated health), and had a high school education or below (61.1%) (Table 1).

Table 1. Characteristics of Study Sample (N=2,823)

	Overall	Changes in eligibility status		
		Unchanged	Changed	
	N (%) or Mean±SD	N (%) or Mean±SD		p-value
Age	60.7±8.2	61.2±8.5	59.1±7.2	<0.001**
Sex				0.261
Male	1289 (45.7)	975 (75.6)	314 (24.4)	
Female	1534 (54.3)	1132 (73.8)	402 (26.2)	
Race				<0.001**
Non-Hispanic Whites	1426 (50.6)	1022 (71.7)	404 (28.3)	
Non-Hispanic Black People	849 (30.1)	652 (76.8)	197 (23.2)	
Hispanics	383 (13.6)	313 (81.7)	70 (18.3)	
Other	160 (5.7)	115 (71.9)	45 (28.1)	
Marital Status				0.901
Married	1090 (38.7)	818 (75.0)	272 (25.0)	
Separated/divorced/widowed	1343 (47.6)	997 (74.2)	346 (25.8)	
Never married	387 (13.7)	289 (74.7)	98 (25.3)	
Education				0.229
High school and below	1726 (61.1)	1269 (73.5)	457 (26.5)	
Some college	825 (29.2)	629 (76.2)	196 (23.8)	
College and above	272 (9.6)	209 (76.8)	63 (23.2)	
Household Income				0.178
<25th	708 (25.1)	542 (76.6)	166 (23.4)	
25-50th	705 (25.0)	509 (72.2)	196 (27.8)	
50-75th	708 (25.1)	521 (73.6)	187 (26.4)	
>75th	702 (24.9)	535 (76.2)	167 (23.8)	

	Overall	Changes in eligibility status		
		Unchanged	Changed	
	N (%) or Mean±SD	N (%) or Mean±SD		p-value
Insurance				0.22
Yes	2398 (84.9)	306 (77.1)	91 (22.9)	
No	397 (14.1)	1779 (74.2)	619 (25.8)	
Self-Rated Health				0.503
Excellent	118 (4.2)	92 (78.0)	26 (22.0)	
Very good	515 (18.3)	395 (76.7)	120 (23.3)	
Good	954 (33.8)	708 (74.2)	246 (25.8)	
Fair	876 (31.1)	653 (74.5)	223 (25.5)	
Poor	356 (12.6)	256 (71.9)	100 (28.1)	

*$p < 0.05$

**
$p < 0.01$

Rates of Screening Eligibility

The results revealed substantial increases in screening eligibility rates for all racial and ethnic groups when comparing the original and revised guidelines (Whites: 35.1% vs. 63.4%; Black people: 13.3% vs. 36.5%; Hispanics: 12.0% vs. 30.3%; Other: 23.8% vs. 51.9%). The largest within group increase was observed among Black people (174%), followed by Hispanics (152%), those in the other category (118%), and Whites (80.8%) (Table 2). This indicates that the revised guidelines successfully expanded eligibility for a significant number of individuals, particularly among Black people and Hispanics. Nevertheless, when considering the percentage increase in eligible individuals for LDCT screening relative to the total population, the percentage of changes in eligibility was highest among Whites who smoke (28.3%), followed by individuals in the “other” category (28.1%), Black people who smoke

(23.2%), and Hispanics who smoke (18.3%). These differences in changes in eligibility were all statistically significant ($p<0.001$).

Table 2. Original USPSTF Guidelines (2013) Compared to Revised USPSTF Guidelines (2021)

	Pack-year	Met original eligibility	Met revised eligibility	% increase in el
Race	Mean±SD	N (%)	N (%)	
Non-Hispanic Whites (n=1426)	31.0±21.9	500 (35.1)	904 (63.4)	80.8%
Non-Hispanic Black People (n=849)	19.3±16.2	113 (13.3)	310 (36.5)	174%
Hispanics (n=383)	17.6±16.3	46 (12.0)	116 (30.3)	152%
Other (n=160)	28.0±22.3	38 (23.8)	83 (51.9)	118%

Factors Associated with Changes in Screening Eligibility

Next, we conducted binary logistic regression analyses to examine further factors associated with changes in screening eligibility status (Table 3). After adjusting for demographic variables, the results revealed the impact of the revised screening guidelines on disparities in eligibility among different racial and ethnic groups. Black people who smoke had a 29% lower likelihood of qualifying for LDCT screening (OR=0.71, p=0.001) than Whites who smoke. Similarly, Hispanics who smoke had even lower odds of eligibility, with a 46% reduced likelihood (OR=0.54, $p<0.001$) compared to Whites who smoke. These findings highlight the persistent disparities in screening eligibility between racial and ethnic groups, even after accounting for demographic factors. Additionally, several demographic factors were found to influence the likelihood of experiencing a change in eligibility status. For example, older respondents (OR = 0.96, $p<0.001$) or uninsured (OR = 0.76, p = 0.039) were less likely to transition from not eligible to eligible status.

Table 3. Changed Status of Screening Eligibility by Race: Binary Logistic Regression Results

	Changes in eligibility status (changed vs. Unchanged)	
	OR (95% CI)	**p-value**
Race		
Non-Hispanic Whites	Reference	
Non-Hispanic Black People	0.71 (0.57-0.87)	0.001**
Hispanics	0.54 (0.40-0.73)	<0.001**
Other	0.88 (0.61-1.29)	0.519
Age	0.96 (0.95-0.97)	<0.001**
Sex		
Male	0.96 (0.80-1.14)	0.621
Female	Reference	
Marital Status		
Married	Reference	
Separated/divorced/widowed	1.04 (0.85-1.28)	0.709
Never married	0.95 (0.70-1.27)	0.714
Education		
High school and below	1.22 (0.88-1.67)	0.231
Some college	0.98 (0.70-1.37)	0.901
College and above	Reference	
Household Income		
<25th	1.08 (0.81-1.44)	0.615
25-50th	1.40 (1.07-1.84)	0.014*
50-75th	1.22 (0.95-1.57)	0.128
>75th	Reference	
Insurance		

	Changes in eligibility status (changed vs. Unchanged)	
	OR (95% CI)	**p-value**
Yes	Reference	
No	0.76 (0.58-0.99)	0.039*
Self-Reported Health		
Excellent	Reference	
Very good	1.02 (0.63-1.67)	0.931
Good	1.17 (0.74-1.87)	0.502
Fair	1.15 (0.72-1.84)	0.572
Poor	1.23 (0.74-2.04)	0.428

*
$p < 0.05$

**
$p < 0.01$

Do the Revised Guidelines Reduce Screening Inequities?

To determine whether the revised guidelines reduce observed racial/ethnic disparities in screening eligibility compared to the original guidelines, it is necessary to compare the absolute differences in screening eligibility between Whites and other racial/ethnic groups under both guidelines (see Figure 1). Under the original guidelines, the study found that Whites who smoke had a screening eligibility rate of 35.1%, while individuals in the “other” category had a rate of 23.8%, Black people who smoke had a rate of 13.3%, and Hispanics who smoke had a rate of 12%. This resulted in absolute differences of 11.3% between Whites and the “other” category, 21.8% between Whites and Black people who smoke, and 23.1% between Whites and Hispanics who smoke.

When comparing the revised guidelines, the study reported a screening eligibility rate of 63.4% for Whites who smoke, 51.9% for individuals in the “other” category, 36.5% for Black people who smoke, and 30.3% for Hispanics who smoke. This resulted in absolute differences of 11.5% between Whites and the “other” category, 26.9% between Whites and Black people who smoke, and 33.1% between

Whites and Hispanics who smoke. Based on the absolute differences in screening eligibility rates between Whites and other racial/ethnic groups, the disparities widened under the new guidelines, particularly with larger absolute differences in screening eligibility observed between Whites, Black people, and Hispanics.

Discussion

This study aimed to assess the impact of changes in LDCT lung cancer screening criteria on racial and ethnic disparities in screening eligibility. Specifically, the study compared Black people and Hispanics who smoke to Whites who smoke using the original and revised USPSTF screening guidelines. Our study findings suggest positive outcomes and ongoing disparities regarding the changes in LDCT lung cancer screening eligibility criteria. Given the reduction in the minimum age and smoking pack-years of revised screening eligibility, the percentage of individuals meeting the eligibility criteria increased among all racial/ethnic groups. Our study findings were consistent with prior research, showing that expanding screening guidelines resulted in more individuals eligible for screening among Black people[24,34,37, 38, 39] and Hispanics.[37,39] The increased eligibility rates among Black people and Hispanics indicate successful expansion of access to screening for these populations, addressing existing disparities.

In contrast to prior research, our study conducted multivariate analyses to examine the likelihood of changes in screening eligibility among racial and ethnic groups by controlling for demographic factors. The likelihood of individuals becoming qualified for LDCT screening remains relatively low among Black people and Hispanics compared to Whites who smoke. Indeed, given the larger absolute differences in screening eligibility rates between Whites, Black people, and Hispanics, study results suggest that disparities in screening eligibility may have actually widened under the new guidelines.

Implications

The study's findings highlight the promise and limitations of changes in eligibility criteria to address disparities in access to LDCT lung cancer screening among Black people and Hispanics who smoke. Increasing eligibility rates for these populations is a step toward reducing healthcare disparities. By broadening the criteria, more individuals from historically underserved communities can benefit from early detection and timely treatment, improving lung cancer outcomes and reducing health disparities. However, the study also emphasizes persistent disparities despite the revised guidelines. A large and consistent body of research has documented that, on average, Black people and Hispanics who smoke fewer cigarettes per day than Whites.[40, 41, 42, 43] As such, continuing to base screening

eligibility on the number of pack years smoked will continue to disadvantage low-frequency people who smoke.[44]

Achieving equitable access to screening for all populations requires additional research to identify appropriate ways to identify high-risk people who smoke to ensure fairness and inclusivity. Prior studies have suggested the benefit of risk prediction modeling as an alternative approach that defines screening eligibility based on individuals' risk. For example, numerous risk prediction models have been developed not only based on demographic factors (e.g., age and smoking history) but also include clinical risk factors associated with lung cancer (e.g., personal and family history of cancer, history of lung-related diseases or conditions) and biomarker information (e.g., germline mutations or protein-based biomarkers).[45,46] The selection of individuals for lung cancer screening based on personal lung cancer risk has outperformed the eligibility criteria used in the screening trials.[47,48]

Interestingly, socioeconomic factors, such as household income and insurance status, played a significant role in eligibility changes. Lower income and lack of insurance are additional barriers to meeting the screening criteria. Recognizing these socioeconomic factors when developing and implementing screening guidelines is crucial, as systemic barriers can disproportionately affect specific populations.[49] Efforts should be made to improve healthcare access and address these barriers to promote equitable screening eligibility.[50]

The results of this study have significant policy and healthcare intervention implications. Policymakers and healthcare professionals should consider these findings when refining eligibility determination for lung cancer screening. Regularly examining the differential impact of screening guidelines is essential to address disparities and ensure equitable access to healthcare services. However, expanding screening eligibility might not guarantee increases in screening uptake among individuals at elevated risk of lung cancer. Prioritizing equitable access to screening for all populations, particularly historically underserved groups, is important and needed.[50,51] This requires addressing barriers created by the existing screening guidelines, socioeconomic barriers, improving healthcare access, and implementing targeted interventions to reduce disparities and enhance the inclusivity and effectiveness of screening programs.[52,53] The study underscores the importance of ongoing evaluation and refinement of eligibility criteria to optimize screening effectiveness and promote equity in healthcare. Guidelines can better serve at-risk populations by adapting eligibility criteria based on emerging research.

Limitations

Several limitations should be acknowledged in this study. Firstly, the reliance on self-reported data in the Health and Retirement Study introduces the possibility of self-reporting and recall biases. Such biases may impact key variables, including smoking behavior and history, crucial for determining eligibility for LDCT screening. While no previous research indicates racial differences in the willingness and accuracy of reporting long-term smoking, it is important to consider the potential influence of self-reporting bias on the findings. Secondly, due to significant missing data on smoking history (i.e., number of cigarettes smoked and years of quitting smoking) among formal people who smoke, the study only included people who currently smoke in the analysis. Therefore, the impact of the revised guidelines on racial disparities in screening eligibility cannot be generalized to former people who smoke within the study.

Additionally, the study utilized data from the Health and Retirement Study conducted in 2016, which may not fully represent the current population or reflect recent changes in smoking behaviors. Lastly, as with any observational study, causality cannot be determined due to the inherent limitations of the study design. Despite these limitations, this study provides valuable insights into the impact of changes in eligibility criteria for LDCT lung cancer screening. It highlights the need for further research and interventions to address disparities and ensure equitable access to screening for all populations.

Conclusions

In conclusion, while the revised USPTF lung cancer screening guidelines have increased screening eligibility rates, they have also widened the disparities in eligibility between racial and ethnic groups. Addressing these disparities requires targeted interventions, addressing socioeconomic and access-related barriers, and ongoing efforts to prioritize equity in lung cancer screening. By doing so, healthcare systems can work towards reducing disparities and ensuring that all populations, regardless of race or ethnicity, have equal access to life-saving screening and timely treatment for lung cancer.

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